

Last Time:

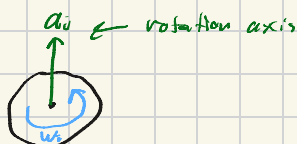
- TVLQR tracking

Today:

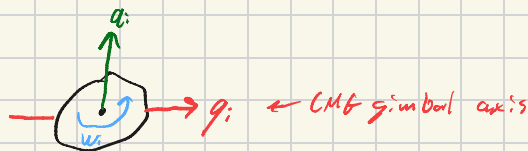
- Control Moment Gyros

CMGs vs Reaction Wheels

- A reaction wheel has a fixed ${}^B a_i$ and changes its speed / momentum w_i



- A CMG typically has a fixed momentum (speed) w_i but changes its axis ${}^B a_i$ by rotating about some gimbal axis ${}^B g_i$:



- This is a fixed-speed single-gimbal CMG. There are also double-gimbal and variable-speed CMGs, but we'll focus on the SGCMG case.

- CMG momentum: ${}^B h_i = w_i a_i$

- CMG torque ${}^B \tau_i = -\dot{{}^B h_i} = -w_i \dot{a}_i = -w_i (g_i \times a_i) u_i$

$\leftarrow$ gimbal rate

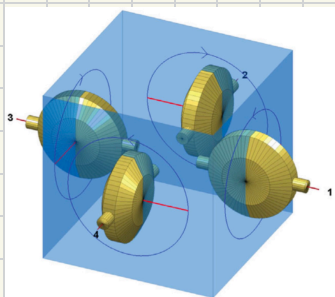
- Note: the torque applied by the CMG on the spacecraft doesn't come from the gimbal motor (or the motor keeping the wheel spinning).

⇒ CMBs can supply very large torques with relatively small motors and low power.

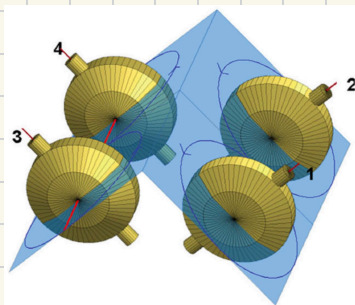
- This is called "torque amplification"
- CMBs are generally used on "high agility" spacecraft that need to execute very fast slow maneuvers.

CMB Arrays:

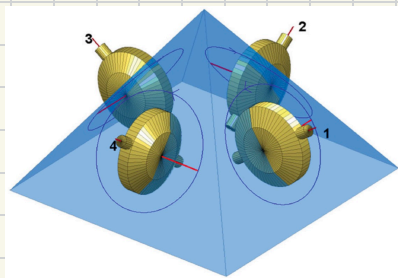
- Typically, spacecraft have at least 4 CMBs. They can be arranged in many ways:



"box 90"

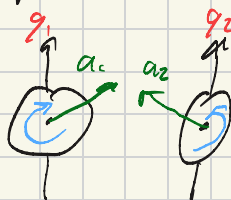


"roof"



"pyramid"

- From a control perspective, the simplest configuration is a "scissored pair"



- By aligning their gimbal axes and synchronizing their gimbal angles, a scissored pair acts like a "virtual reaction wheel"

- Has CMG's torque advantage compared to reaction wheels, but gives up some performance vs. other configurations.
-

Spaccraft + Dynamics with CMGs:

- We'll derive a simplified model good for control design
- We'll assume the inertia/momentum is negligible, so that we can just account for the spinning wheel.
- As usual, we start with the gyrostad equation:

$$J\ddot{\alpha} + \dot{p} + \omega \times (J\omega + p) = \tau$$

- The total rotor momentum for m CMGs is:

$$p = \sum_{i=1}^m w_i a_i$$

- Differentiate:

$$\dot{p} = \sum_{i=1}^m w_i \dot{a}_i = \sum_{i=1}^m w_i (g_i \times a_i) u_i$$

- We can write a wheel Jacobian

$$\tau_w = -\dot{p} = \underbrace{\begin{bmatrix} -w_1 (g_1 \times a_1) & -w_2 (g_2 \times a_2) & \dots & -w_m (g_m \times a_m) \end{bmatrix}}_{B_w} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}$$

- We can stack all of this into a Gg dynamics function:

$$\underbrace{\begin{bmatrix} \dot{q} \\ \dot{\omega} \\ \dot{a}_1 \\ \vdots \\ \dot{a}_m \end{bmatrix}}_{\dot{x}} = \begin{bmatrix} \frac{1}{2} G(q) \omega \\ -J^{-1} \left[\omega \times (J\omega + \sum w_i a_i) - B_w(x) u \right] \\ (g_1 \times a_1) u_1 \\ \vdots \\ (g_m \times a_m) u_m \end{bmatrix} \underbrace{\hspace{10em}}_{f(x, u)}$$

- Note: we've chosen to keep track of the unit vector wheel axes in the state. Instead, it's common to use scalar gimbal angles.

CMG Steering Laws:

- CMGs are usually used with a "controller" that outputs torques and a "steering law" that converts torques to gimbal rates.
- We'll use a simple PD controller:

$$\tau = -K_p \phi - K_d \omega, \quad \phi = \log(L^+(\bar{q})q)$$

reference attitude

- Using the wheel Jacobian:

$$\begin{bmatrix} \tau \end{bmatrix} = \begin{bmatrix} B_w(x) \end{bmatrix} \begin{bmatrix} u \end{bmatrix}$$

- For a given τ , u is not unique
- Like with reaction wheels, we can use a pseudo inverse:

$$\left. \begin{array}{l} \min_u \frac{1}{2} u^T u \\ \text{s.t. } Bu = \tau \end{array} \right\} \Rightarrow u = B^T (BB^T)^{-1} \tau$$

must be full rank

- Unlike with reaction wheels, $B_w(x)$ is a function of the state and is time varying. It can easily become singular.
- Simple hack: "regularization"

$$u = B^T (BB^T + \epsilon I)^{-1} \tau$$

small positive constant

- Regularization guarantees we can solve for u , but it perturbs the solution.
- Equivalent to relaxing the constraint into a penalty:

$$\min_u \frac{1}{2} u^T u + \frac{1}{2} \frac{1}{\epsilon} (Bu - \tau)^T (Bu - \tau)$$

$$\Rightarrow B_w(x)u \neq \tau$$

- This can cause problems

* Simulation

- In principle, we should be able to steer CMBs around singularities while still exactly satisfying $B_w u = \tau$ by exploiting extra degrees of freedom (assuming we have more than 3 CMBs)
- There are lots of papers on fancy CMB steering laws that try to do this.